Number Theory — problem 6.135

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Fifty numbers $a_1, a_2, \ldots, a_{50}$ are written along a circle ; each of the numbers is $+1$ or -1. You want to find the product of these numbers. You may find the product of three consecutive numbers in one question. How many questions do you need at least ?

Here is a generalization you can work on : Along a circle are written n numbers, each number being $+1$ or -1. Our aim is to find the product of all n numbers. In one question, we can find the product of k successive numbers $a_i \cdots a_{i+k-1}$. Here $a_{n+1} = a_1$, and so on. How many questions $q(n, k)$ are necessary to find the product ?